UNC Charlotte

ECGR 3120 Electromagnetics

Magnetically Steered Cardiac Ablation

Milestone 5: System Integration and Engineering Recommendation

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Input status: The steering-to-seating-depth relation and 20 mm^3 proxy target are illustrative. Coil winding, cooling, return boundary, and electrode-drive assumptions are not official specifications.

Abstract

This report integrates the existing Milestone 1–4 calculations without changing their source files or results. The Milestone 1 Helmholtz field model sets field and coil current; Milestone 2 predicts catheter angle; an explicitly illustrative angle-error relation maps steering error to seating depth; the saved Milestone 4 refinement data supplies tissue impedance, myocardium power share, and a thresholded power-density volume proxy. Milestone 3 estimates coil dissipation, temperature rise, and attractive mounting force. Under the illustrative assumptions, 80 mT is the lowest evaluated field meeting a 20 mm^3 proxy target across the sweep and observed grid envelope. This is not a clinical recommendation: the seating-depth relation is not official, the cooling target is exceeded at 80 mT, and the proxy is not a predicted thermal lesion.

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1 Objective and integrated system

Milestone 5 connects the previous four standalone models into one system-level study:

$$\begin{aligned}\text{coil current} &\rightarrow \text{magnetic field} \rightarrow \text{magnetic torque}\\ &\rightarrow \text{achieved angle} \rightarrow \text{contact depth}\\ &\rightarrow \text{tissue electrical metrics and proxy volume.}\end{aligned}$$

The evaluation covers 40, 80, and 160 mT center fields and commanded angles from 0° through 90° in 5° steps. Coil power and mounting force are calculated for each field. Existing M1–M3 functions are reused. The M4 grid-refinement CSV is interpolated in contact depth, avoiding a costly rerun of its finite-volume sweeps.

2 Inputs and assumptions

Table 1: Input status for the integrated model.

Input	Value	Status
Helmholtz coil radius and spacing	0.35 m each	Handout geometry used by M1
Tip magnetic moment	approximately $0.013\,\mathrm{A\,m^2}$	Approximate handout value
Shaft stiffness	9.93×10^{-4} N m rad^{-1}	M2 calibration from illustrative 80 mT, 60ř command, 30ř tip case; not measured
Center field cases	40 mT, 80 mT, 160 mT	Requested comparison targets, not verified hardware settings
Turns per coil; conductor radius	1000; 4 mm	Illustrative M3 example
Copper resistivity	$1.724 \times 10^{-8}\ \Omega\,\mathrm{m}$	Approximate at 20řC; material/temperature not confirmed
Convection coefficient; pack fill	$5\,\mathrm{W\,m^{-2}\,K^{-1}}$; 0.65	Illustrative M3 assumptions
Electrode radius; tissue conductivities	2 mm; 0.54 S/m myocardium; 0.70 S/m blood	M4 handout inputs
Outer return radius; electrode drive	20 mm; 1 V	Illustrative M4 assumptions
Power-density threshold	$5.0 \times 10^4\ \mathrm{W\,m^{-3}}$	Illustrative electrical proxy cutoff, not an injury criterion
Steering-error/depth relation	Linear, configurable illustrative model	No official relationship found; equation is stated below
Minimum proxy-volume target	$20\,\mathrm{mm^3}$	Selected illustrative comparison target, not a medical requirement
Maximum temperature rise	40 K	Illustrative M3 design target

The starter files and prior reports do not specify an official relationship between steering-angle error and electrode seating depth. To permit an end-to-end sensitivity study, this report uses a clearly labeled linear scenario relation:

$$d = \mathrm{clip}_{[0,d_{\max}]}\left[d_{\max}\left(1 - \frac{|e|}{e_{\max}}\right)\right], \qquad d_{\max} = 4\,\mathrm{mm}, \quad e_{\max} = 90^\circ,$$

where $e = \theta_c - \theta_t$. Zero error maps to maximum seating and an error of 90° or more maps to zero seating. This is not a measured catheter-geometry law. The depth is clipped to M4's validated 0–4 mm lookup range, and uncertainty in this mapping is not quantified.

3 Governing equations and numerical method

The M1 field model evaluates the two-coil Helmholtz field at the origin with the existing midpoint-discretized Biot–Savart function. Ampere-turns are scaled to each requested center field. The per-turn current is $I = NI/N$, using the illustrative 1000-turn M3 example.

The existing M2 solver calculates the equilibrium from

$$mB\sin(\theta_c - \theta_t) = k\theta_t,$$

where m is the tip moment, B is the field magnitude, θ_c is the commanded angle, and θ_t is the achieved angle. Angles are evaluated in radians. The shaft stiffness is the prior M2 calibration, not an independent measurement.

For M3, the same turn count, current, coil geometry, and resistivity are used to calculate

$$L = N(2\pi R), \qquad R_{\text{coil}} = \rho\frac{L}{A}, \qquad P_{\text{coil}} = I^2 R_{\text{coil}}.$$

The attractive force is computed using the existing discretized circular-loop integral for the coaxial coils; ampere-turns are not substituted for current in the Joule-heating equation. The lumped free-convection rise is $\Delta T = P_{\text{coil}}/(hA_{\text{exposed}})$, using the illustrative M3 winding-pack estimate.

For each derived contact depth, the script interpolates the saved M4 values at 0, 1, 2, 3, and 4 mm. The 6-cells-per-radius result is the sampled nominal. The minimum and maximum of the independently interpolated 3-, 4-, and 6-cell results are carried separately for impedance, myocardium power fraction, and thresholded volume. No impedance error is applied to volume. M4 electrical and myocardium powers below refer to its illustrative 1 V electrode drive, not to the navigation coils.

4 Integrated comparison

At a 60° command, the achieved angles reproduce the previous M2 results. The selected depth mapping makes the 90° command the limiting proxy-volume case for all three fields.

Table 2: Field-strength and system-cost comparison over the 0–90ř command sweep.

Field (mT)	Tip at 60ř (deg)	Depth at 90ř (mm)	Minimum proxy (nominal; grid range) (mm^3)	Current/coil (A)	Ampere-turns/coil (A-turn)	Power/coil (W)	Pair power (W)	Force (N)	Rise (K)
40	19.488	1.190	16.428; 11.737–16.428	15.569	15,569.5	182.837	365.674	174.137	14.949
80	30.000	1.934	24.470; 20.550–24.470	31.139	31,139.0	731.348	1,462.697	696.547	59.795
160	40.351	2.667	32.494; 26.864–32.500	62.278	62,278.0	2,925.393	5,850.786	2,786.189	239.182

At the limiting 90° command, the interpolated M4 nominal values and grid ranges are:

Table 3: Tissue-model results at the 90ř commanded-angle endpoint.

Field (mT)	Tip angle (deg)	Error (deg)	Depth (mm)	Impedance, nominal [range] (Ω)	Myocardium share, nominal [range] (%)	Proxy volume, nominal [range] (mm^3)
40	26.782	63.218	1.190	55.657 [52.107–55.657]	33.334 [30.737–33.378]	16.428 [11.737–16.428]
80	43.513	46.487	1.934	57.090 [53.809–57.090]	42.715 [42.503–42.718]	24.470 [20.550–24.470]
160	60.000	30.000	2.667	58.645 [54.815–58.645]	53.225 [49.658–53.225]	32.494 [26.864–32.500]

The M5 80 mT calculation gives 731.348 W per coil and 696.547 N, slightly different from the prior M3 illustrative outputs of 731.378 W per coil and 696.575 N. M3 used the analytical M1 ampere-turn estimate; M5 scales ampere-turns against the existing 512-segment Biot–Savart center field (80.001 606 mT for that analytical estimate) to produce the requested 80 mT case. The approximately 0.004% difference is a field-scaling convention, not a change to previous milestone results.

For the recommended 80 mT case at the limiting command, the M4 model gives 17.516 mW tissue electrical input and 7.482 mW deposited in myocardium at the nominal 6-cell power fraction. This is distinct from the approximately 731 W dissipated in each navigation coil. All 57 command/field combinations, with separate grid ranges for the M4 metrics, are in `m5_system_sweep.csv`.

5 System response and design tradeoffs

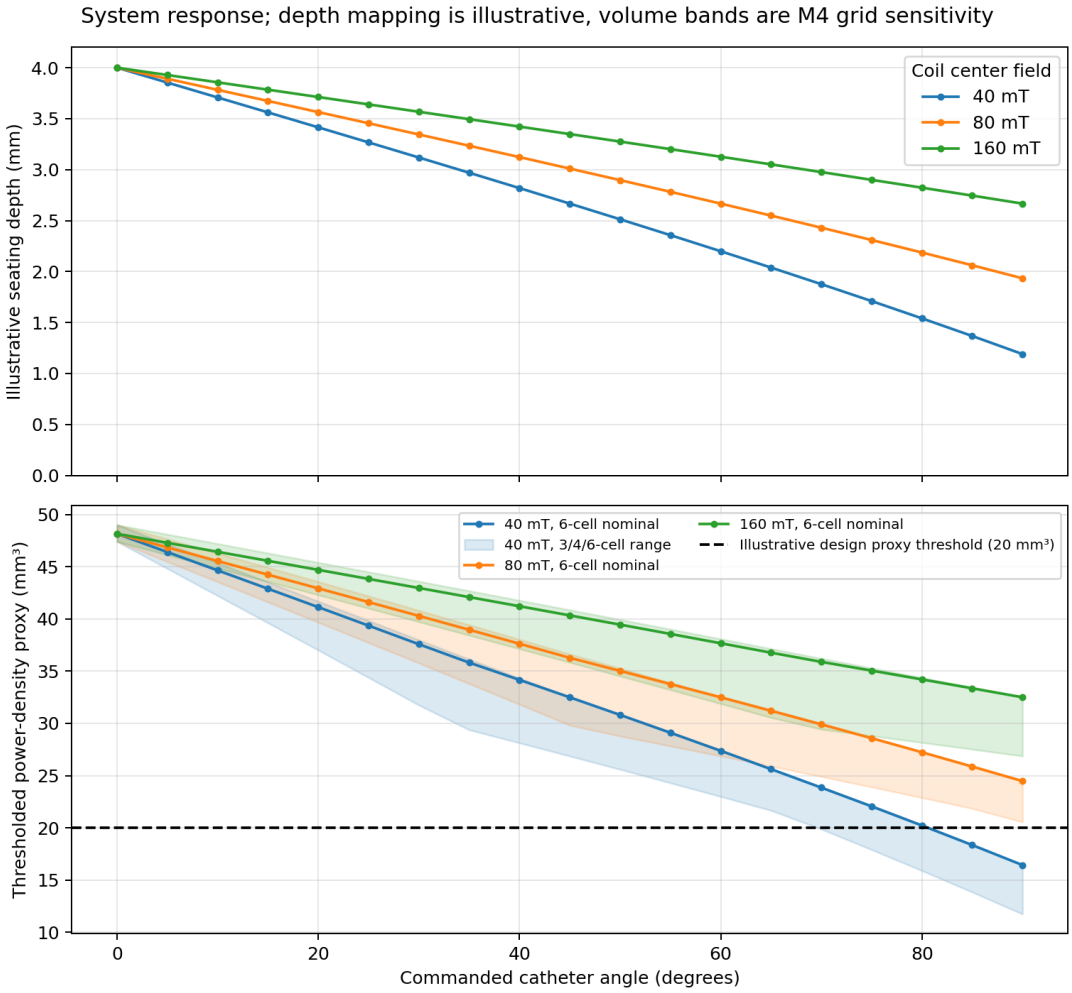


Figure 1: Command angle versus the illustrative seating depth and M4 thresholded power-density volume proxy. Bands on the proxy show the observed 3/4/6-cell M4 grid range, not a confidence interval. The proxy is not a clinical or thermal lesion prediction.

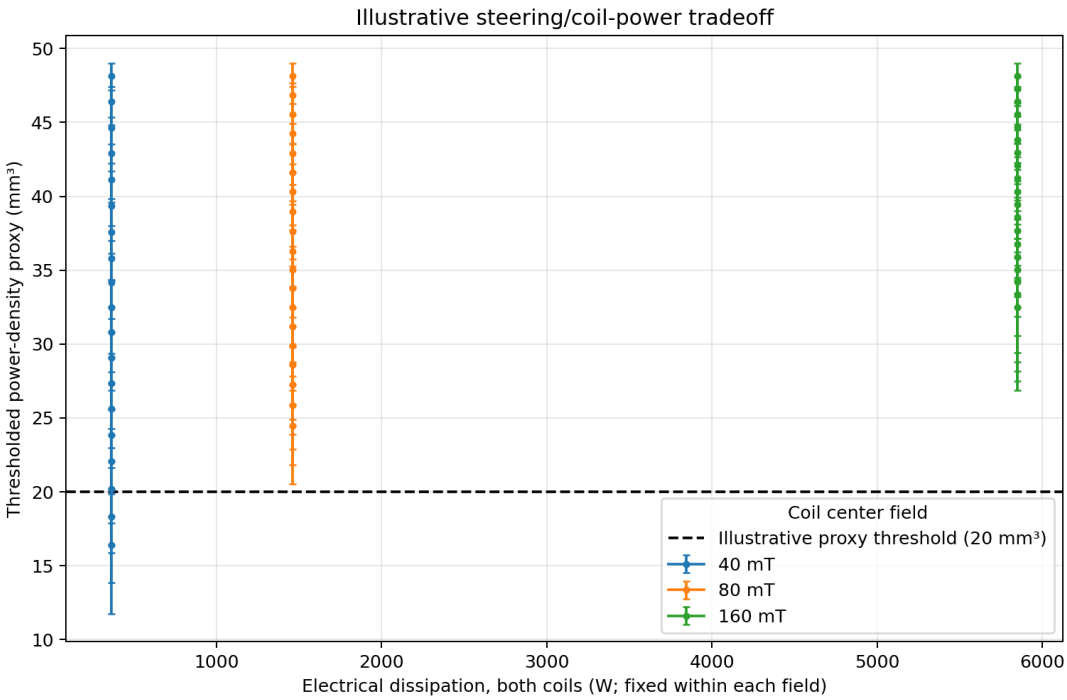


Figure 2: Thresholded M4 proxy volume versus total dissipation in both navigation coils as command angle varies. Error bars show interpolated M4 grid sensitivity only.

The first figure shows that, under the assumed depth mapping, higher fields reduce angular error and preserve more seating depth at high commands. The second figure makes the electrical cost explicit: at fixed coil geometry and turn count, field and current scale together, while power and mutual force scale approximately with current squared. Both plots use the simulated values; no graph points were manually selected.

6 Numerical uncertainty and limitations

The carried M4 ranges are observed finite-grid envelopes, not statistical confidence intervals or guaranteed error bounds. At the recommended field's limiting 90° command, the interpolated ranges are:

$$R = 53.809\text{–}57.090\ \Omega \quad \text{(6-cell nominal 57.090 } \Omega\text{)},$$

$$f_{\text{myo}} = 42.503\text{–}42.718\% \quad \text{(6-cell nominal 42.715\%)},$$

$$V_{\text{proxy}} = 20.550\text{–}24.470\ \text{mm}^3 \quad \text{(6-cell nominal 24.470 mm}^3\text{)}.$$

These are separate sensitivities. The impedance interval is not used as a proxy-volume error estimate. The volume is a cell-counted threshold result; refinement stopped at 6 cells/radius. Linear interpolation between five depth samples adds another approximation. The seating-depth relation, 1 V drive, 20 mm return boundary, and power-density cutoff are outside these grid envelopes. A reliable total uncertainty bound is therefore not established.

The thresholded M4 volume is an electrical power-density proxy, not a thermally predicted or clinically validated lesion. No bioheat, perfusion, transient temperature, or tissue-damage model is

included. The M2 spring constant was inferred from an approximate baseline, and the M3 current, turns, wire, cooling, winding pack, and allowable temperature rise are illustrative. Field uniformity, actual spatial catheter position, dynamic steering, contact mechanics, and structural safety factors are not included.

7 Validation

Table 4: Integrated model checks.

Check	Result
M1 Biot–Savart field reproduces each requested center-field target	PASS
M2 magnetic/spring torque residual across the sweep is $\leq 10^{-12}$ N m	PASS
Illustrative depth remains within the M4 0–4 mm domain	PASS
Achieved angle increases with field at fixed command	PASS
M3 coil power follows current-squared scaling	PASS
M3 full-loop attractive force follows current-squared scaling	PASS
Each nominal M4 metric is enclosed by the carried 3/4/6-cell range	PASS
Local regression suite, `tests/test_m5_system.py`	3 passed
Professor-provided M5 and M1–M4 acceptance tests	Unavailable

The local integration tests exercise the existing M1–M4 interfaces and are not professor acceptance tests. No earlier milestone report, source, figure, or numerical result was changed for this integration.

8 Final engineering recommendation

For an illustrative system-screening objective, use a minimum thresholded proxy volume of 20 mm^3 at all 19 sampled commands from 0° to 90° in 5° increments. This is a comparison target selected for this model, not a medical, clinical, or professor-provided requirement.

At 40 mT, the worst-command nominal proxy is 16.428 mm^3, with an observed lower grid value of 11.737 mm^3, so it does not meet the selected target. At 80 mT, the worst-command nominal is 24.470 mm^3 and the lower observed grid value is 20.550 mm^3. This makes **80 mT the lowest evaluated field meeting the target across those samples and the observed M4 grid envelope**. The 160 mT case also meets it, but uses four times the coil power and mounting force of 80 mT for this fixed geometry and winding.

The 80 mT case requires approximately 31.139 A and 31,139 A · turn per coil. It dissipates 731.348 W per coil, or 1.463 kW for the pair, and produces an estimated attractive mounting load of 696.547 N. Its free-convection estimate is 59.795 K, above the illustrative 40 K target. Therefore, 80 mT is a **conditional model recommendation**, not a verified hardware or clinical recommendation: cooling and supports require additional design and verification.

The 40 mT alternative has lower power, mounting force, and calculated temperature rise, and meets the illustrative thermal target, but misses the proxy target across the sampled command range. The 160 mT alternative yields a larger proxy but raises the modeled temperature rise to approximately 239 K, pair dissipation to 5.851 kW, and mounting force to 2.786 kN. Under the

selected proxy target, 80 mT is the least costly evaluated option, although its cooling shortfall remains a design concern.

9 How I Would Explain Milestone 5 to My Professor

I connected the previous milestones without changing their calculations. The Helmholtz model converts ampere-turns into field, the steering model balances magnetic torque against the calibrated shaft spring, and the coil model calculates the electrical and mechanical cost of each field. Because the starter files did not define how angular error changes electrode seating, I added a clearly labeled, configurable illustrative mapping and passed its depth into the saved M4 grid results. The 80 mT case is the lowest of the three tested fields that clears my illustrative 20 mm^3 electrical proxy target over the command sweep and the observed grid range. Its coil heating and roughly 697 N attraction show why this is only a conditional model recommendation: the cooling and mechanical design still need real project parameters and verification.